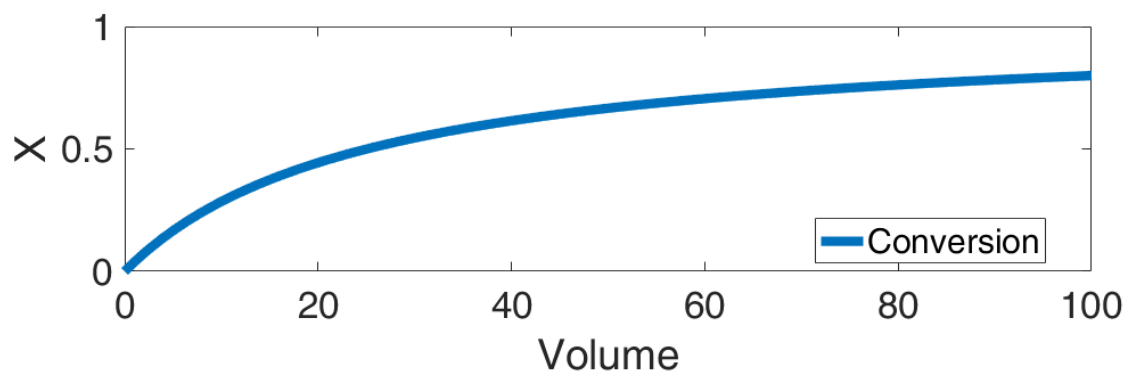
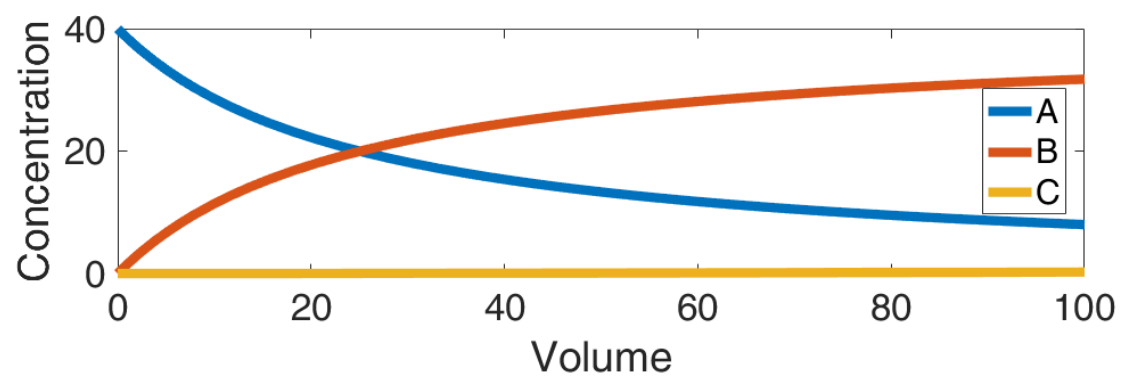




```

clc;
clear all;

L=10;
A=0.01;
caf=40;

v=1;
vr=0;
k1=0.001;
k2=0.0001;

%Outlet boundary conditions
cal=caf;
cbl=0;
ccl=0;

cend(:,:)=zeros;

for i=1:1:5

    cend(i,1)=cal;
    cend(i,2)=cbl;
    cend(i,3)=ccl;
    cend(i,4)=i;

    %Inlet boundary conditions
    cao= (v*caf + vr*cal)/(v+vr);
    cbo= (vr*cbl)/(v+vr);
    cco= (vr*ccl)/(v+vr);

    volspan=[0 A*L*1000]; %total volume range

    %Initial condition for the solver
    cinit=[cao; cbo; cco];

    [vol,c] = ode45(@che321_func,volspan,cinit);

    %Change cal,cbl,ccl in every iteration
    %Size of vol matrix
    s=size(vol);
    cal=c(s(1,1),1);
    cbl=c(s(1,1),2);
    ccl=c(s(1,1),3);

```

end

```
graph=plot(cend(:,4),cend(:,1),cend(:,4),cend(:,2),cend(:,4),cend(:,3));
    set(graph,'LineWidth',5)
    xlabel('iteration');
    ylabel('Concentration');
    legend('A','B','C','Location','Best')
    set(gca,'FontSize',20)

%     subplot(2,1,1)
%     graph2=plot(vol,c(:,1),vol,c(:,2),vol,c(:,3));
%     set(graph2,'LineWidth',5)
%     xlabel('Volume');
%     ylabel('Concentration');
%     legend('A','B','C','Location','Best')
%     set(gca,'FontSize',20)
%
%
%     subplot(2,1,2)
%     graph2=plot(vol,(caf-c(:,1))/caf);
%     set(graph2,'LineWidth',5)
%     xlabel('Volume');
%     ylabel('X');
%     legend('Conversion','Location','Best')
%     set(gca,'FontSize',20)
```

```
function dcdvol = che321_func(vol,c)

v=1;
vr=0;
k1=0.001;
k2=0.0001;

dcdvol=[-k1*c(1)*c(1)/(v+vr); (k1*c(1)*c(1)-
k2*c(2))/(v+vr); k2*c(2)/(v+vr)];
```